

Contact

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Interests

Algebraic geometry
Arithmetic geometry
Complex geometry
Arakelov theory
Intersection theory
Calabi–Yau manifolds
Hodge theory
Analytic torsion
Enumerative mirror
symmetry
Singularities

About me

I am a Swedish
mathematician with
extensive international
research experience,
and a strong interest
in collaboration,
communication and
education. Outside
mathematics, I enjoy
language learning and
cultural travel.

Languages

Swedish ●●●●

English ●●●●

French ●●●●

German ●●●●

Chinese ●●●●

Japanese ●●●●

Polish ●●●●

Research Summary

My research grows out of arithmetic geometry, where algebraic geometry, number theory, and complex analysis meet. This area combines tools from Grothendieck's algebraic geometry with ideas from complex differential geometry, and I use this interaction to study structural invariants of algebraic varieties. In recent years, these methods have naturally connected my work to mathematical aspects of mirror symmetry — a conjectural framework originating in string theory that continues to influence several areas of modern geometry — as well as to questions about singularities. Across these topics, my work studies how algebraic geometry, and in particular intersection theory, serves as a meeting point for arithmetic ideas, complex geometric methods, Hodge theory, singularities, and mirror symmetry.

Select Significant Publications

- 2024 **Deligne-Riemann-Roch and intersection bundles**
J. Éc. polytech. Math. 11 (2024), 247–361
With Gerard Freixas i Montplet.
- 2022 **On genus one mirror symmetry in higher dimensions and the BCOV conjectures**
Forum Math. Pi 10(e19) (2022), 1–53
With Gerard Freixas i Montplet, Christophe Mourougane.
- 2022 **Complex Chern–Simons bundles in the relative setting**
Astérisque (Accepted for publication)
With Gerard Freixas i Montplet, Richard Wentworth.
- 2021 **BCOV invariants of Calabi–Yau manifolds and degenerations of Hodge structures**
Duke Mathematical Journal 170(3) (2021), 379–454
With Gerard Freixas i Montplet, Christophe Mourougane.
- 2021 **Spaces of norms, determinant of cohomology and Fekete points in non-Archimedean geometry**
Advances in Mathematics 378 (2021), 107501–107501 (Article ID 107501)
With Sébastien Boucksom.

Some Select Research Grants

- 2026 **Research in Paris** Institut Henri Poincaré
2 month invitation.
- 2025 **Vergstiftelsen Postdoc Grant** Postdoc grant
Grant, approx. €200,000, to hire a postdoc for two years, in the project "The hunt for new invariants in the intersection of arithmetic and algebraic".
- 2022–2025 **Vetenskapsrådet Research Project Grant** PI
Approx. €350,000.
- 2009–2011 **Vetenskapsrådet Postdoc Grant** PI
Salary for two years.
- 2024 **G.S. Magnussons Grant** mobility grant
Approx. €30,000.
- 2025 **Svensk-Franska stifelsen** mobility grant
Approx. €6,000.
- 2020 **Knut och Alice Wallenberg Foundation** Visiting Professor Grant
For Gerard Freixas i Montplet, one year visiting professor at Chalmers.
- 2018 **Research in Paris** Institut Henri Poincaré
Two-month funded visit.
- 2014 **Trimester Program Organizer** HIM Bonn
Group-leader: "p-adic methods in Arakelov geometry and Shimura varieties".

Positions

2024–present	Professor (Gothenburg) Department of Mathematical Sciences.	Chalmers University of Technology
2018–2023	Associate Professor (Docent) (Gothenburg) Department of Mathematical Sciences.	Chalmers University of Technology
2014–2018	Assistant Professor (Gothenburg)	Chalmers University of Technology
2017	Visitor MPI January–April.	Max Planck Institute for Mathematics, Bonn
2014	Junior Trimester Program (Organiser) Trimester: Algebraic geometry (p-adic methods in Arakelov geometry and Shimura varieties).	Hausdorff Institute for Mathematics
2013	Postdoctoral Researcher January–December.	Max Planck Institute for Mathematics
2009–2012	Postdoctoral Researcher Teaching assistant 2011–2012.	University of Gothenburg
2008–2009	Postdoctoral Researcher	University of Tokyo
2007–2008	JSPS Research Fellow	University of Tokyo

Education

2003–2008	PhD in Mathematics Thesis: Formule de Lefschetz fonctorielle et applications géométriques, under the supervision of Damian Rössler.	Université d'Orsay, Paris 11, Paris
2000–2003	Undergraduate and master studies in Mathematics University of Gothenburg, Gothenburg + Paris 13, Paris Thesis: Some more or less explicit class field theory. Under the supervision of Peter Hegarty.	

Academic & Organizational Responsibilities

2018–2023	PhD Student Admissions Committee Member of the PhD committee deciding future PhD students.	Chalmers
2019–2023	Programme Board Member Member of the "Programråd" for Samhällsbyggnadsteknik.	Chalmers
2014	Group Leader (Junior Trimester Program) Organized and led the Junior Trimester Program: p-adic methods in Arakelov geometry and Shimura varieties.	Hausdorff Institute for Mathematics
2009–2020	High School Outreach Coordinator Contact person for high school research projects in mathematics.	Gothenburg region
2014–	Main Organizer / Co-Organizer (International Workshops) International workshops N-cube (2015, 2016, 2018, 2025); Freixit 2024; Number theory and dynamical systems (2016); Algebraic Geometry (Bonn, 2018); Workshop: Arithmetic intersection theory and Shimura varieties (2014).	

- 2010– **Thesis Committees & Examination** Paris and Gothenburg
 Rapporteur for Dai Imai (Kyoto 2024); Rapporteur and Member of jury for Dorian Ni (Paris 2022); Member of jury for Bertrand Duma (Paris 2012); Diskutant/Opponent at licentiate defense for Ida Säfström (Gothenburg 2010).
- 2010– **Professional Service and Journal Refereeing** International Journals
 Regular ad-hoc referee for journals including: Inventiones Mathematicae, IMRN, Algebra & Number Theory, Journal of Differential Geometry, JEMS, JHEP, JGP, Selecta Mathematica, and Pacific Journal of Mathematics. Also served as a referee for various international conference proceedings.

Teaching and Pedagogical Qualifications

- Quarter 2 **Algebraic geometry**
- Quarter 3 **Algebraic structures**
- 2016– **Undergraduate** Chalmers
 Preparatory Calculus; Calculus for Civil Engineering; Multivariable Calculus.
- 2016– **Master level** Chalmers, Hangzhou, Paris
 Algebraic Geometry; reading courses in algebraic topology and homological algebra.
- 2016– **PhD level** Graduate school
 K-theory (with M. Goffeng); p-adic methods (with A. Södergren, S. Herrero); reading courses in algebraic number theory, Arakelov theory, Hodge theory, derived geometry, and Gromov–Witten theory.
- 2019, 2020, 2023 **Teaching Award, three times** Chalmers
 Prize for best teacher in the Civil Engineering programme. Nominated for the Chalmers pedagogical prize in 2020.
- 2024–present **Bachelor Thesis Examiner** Chalmers & University of Gothenburg
 Examiner for bachelor theses in mathematics; approximately 80 students and 20 projects every year.
- 2018 **Pedagogical Research Publication** Primus 28 (2018), no. 8
 Communicating to Learn Multivariable Calculus: Students' Blackboard Presentations as a Means for Enhancing Mathematics Learning. With Hans Malmström.
- 2018 **Innovation Grant** Chalmers
 Approx. €70,000 for developing digital teaching tools. I applied for and led the project.
- 2014–2018 **Pedagogical Studies in Higher Education (15 hp)** Chalmers
 Formal qualification in university-level teaching and learning, including student assessment, curriculum design, and learning theories.

Supervision

- Postdocs Yanbo Fang (Current, 2026–present)
- PhD students Tudor Sharpe (Current, 2026–present); Gabriel Abánades Joglar (Current, 2025–present); Anthony Mäkelä (Current, 2025–present, Co-supervisor); Douglas Molin (Completed 2026, Co-supervisor); Mykola Pocheckai (Completed 2024); Jiacheng Xia (Completed 2021, Co-supervisor); Manh Hung Tran (Completed 2020)
- Master students Samuel Plumeyer (Current); Sebastian Jansson (Current); Liu Xudong (2024); Felix Frelot (2022); Robert Antonio Pop Gorea (2021); Magnus Fries (2020); Mattias Lindby (2016)

Recent Selected Seminars

Oct 2026	Genus one mirror symmetry for toric varieties	Pohang, South Korea
Sep 2026	Archimedean heights and bad reduction of Calabi–Yau manifolds Westlake University, Hangzhou, China	
Dec 2025	Algebraic geometry seminar On smooth fillings of Calabi–Yau varieties	Bologna, Italy
March 2025	Séminaire de géométrie "Anniversaire Paul 80" The spectral genus and an inequality for the Milnor number	Paris, France
March 2024	Spring school in Arakelov Geometry and related topics Relative intersection theory and Arakelov geometry, minicourse	Hangzhou, China
April 2024	Algebraic geometry seminar From counting genus one curves to questions about singularities	Stockholm, Sweden
Sep 2023	Masterclass: Arakelov geometry on Shimura varieties Eisenstein series and enumerative geometry	Copenhagen, Denmark
2019–2021	Genus one mirror symmetry and the arithmetic Riemann–Roch theorem Princeton, Marseille, Oxford, Stockholm	

Selected Conferences

May 2026	AI for Mathematics The very confident intern — how I use LLMs in my mathematical research	Gothenburg, Sweden
Sep 2025	Kähler Geometry: Past, Present and Future On a conjecture of Deligne for the determinant of the cohomology	Banff, Canada
May 2025	Hodge theory in a broad sense On smooth fillings of Calabi–Yau families	Stockholm, Sweden
Dec 2024	Around Local Index Theory Intersection bundles and the Cappell–Miller torsion for Riemann surfaces. Mini-course on joint work with Gerard Freixas i Montplet.	Paris, France
Nov 2024	Algebra and Friends Milnor Number Inequalities in Hypersurface Singularities	Borås, Sweden
May 2024	Complex Geometry Workshop On some singularity invariants from analytic torsion	Aarhus, Denmark
Sep 2021	Arakelov geometry intercity conference Genus one mirror symmetry and the arithmetic Riemann–Roch theorem	Regensburg, Germany
2019	Workshop Alkage On an invariant for Calabi–Yau manifolds through analytic torsion	Grenoble

Complete List of Publications

- 2026 **Spectrum of singularities, analytic torsion and Calabi–Yau degenerations**
Preprint
With Gerard Freixas i Montplet.
- 2026 **Base change conductors through intersection theory and quotient singularities**
Documenta Mathematica
With Lars-Halvard Halle, Johannes Nicaise.
- 2025 **The spectral genus of an isolated hypersurface singularity and its relation to the Milnor number and analytic torsion**
Documenta Mathematica
With Gerard Freixas i Montplet.
- 2025 **The Deligne–Riemann–Roch isomorphism** Preprint
With Gerard Freixas i Montplet.
- 2024 **Erratum to: Discriminants and Artin conductors (J. Reine Angew. Math. 712 (2016), 107–121)**
J. Reine Angew. Math. 814 (2024), 283–284 (Erratum (MR4793346))
- 2024 **Deligne–Riemann–Roch and intersection bundles**
J. Éc. polytech. Math. 11 (2024), 247–361
With Gerard Freixas i Montplet.
- 2024 **Genus one mirror symmetry for intersection of two cubics in P^5** Preprint
With Mykola Pocheikai.
- 2022 **Complex Chern–Simons bundles in the relative setting**
Astérisque (Accepted for publication)
With Gerard Freixas i Montplet, Richard Wentworth.
- 2022 **On genus one mirror symmetry in higher dimensions and the BCOV conjectures**
Forum Math. Pi 10(e19) (2022), 1–53
With Gerard Freixas i Montplet, Christophe Mourougane.
- 2021 **Global representation of Segre numbers by Monge–Ampère products**
Mathematische Annalen 380(1–2) (2021), 349–391
With Mats Andersson, Håkan Samuelsson Kalm, Elizabeth Wulcan, Alain Yger.
- 2021 **Spaces of norms, determinant of cohomology and Fekete points in non-Archimedean geometry**
Advances in Mathematics 378 (2021), 107501–107501 (Article ID 107501)
With Sébastien Boucksom.
- 2021 **Nonproper intersection products and generalized cycles**
Eur. J. Math. 7(4) (2021), 1337–1381
With Mats Andersson, Håkan Samuelsson Kalm, Elizabeth Wulcan, Alain Yger.

- 2021 **BCOV invariants of Calabi–Yau manifolds and degenerations of Hodge structures**
Duke Mathematical Journal 170(3) (2021), 379–454
With Gerard Freixas i Montplet, Christophe Mourougane.
- 2020 **Kähler quantisation of vortex moduli** Letters in Mathematical Physics 110(3) (2020), 659–693
With Nuno Romão.
- 2018 **Singularities of metrics on Hodge bundles and their topological invariants**
Algebraic Geometry 5(6) (2018), 742–775
With Gerard Freixas i Montplet, Christophe Mourougane.
- 2016 **Discriminants and Artin conductors**
Journal für die reine und angewandte Mathematik (Crelle) 712 (2016), 107–121
- 2015 **A logarithmic interpretation of Edixhoven's jumps for Jacobians**
Advances in Mathematics 279 (2015), 532–574
With Lars-Halvard Halle, Johannes Nicaise.
- 2013 **Degenerating Riemann surfaces and the Quillen metric**
International Mathematics Research Notices 2013(2) (2013), 347–361
- 2009 **Un isomorphisme de type Deligne–Riemann–Roch**
Comptes Rendus Mathématique 347(19-20) (2009), 1115–1118
- 2008 **On the Brauer–Manin obstruction for zero-cycles on curves**
Acta Arithmetica 135(2) (2008), 99–110
With Victor Scharaschkin.